Tianfang Chang

tfchang.ca@gmail.com | 236-882-3992 | Tianfang | Tianfang | Tianfang

Summary

Research based Software Engineer with hands-on C++/Python experience in distributed routing, congestion control, and PyTorch-based deep-learning modeling. Built and evaluated ns-3 prototypes on ISP-grade topologies and co-authored papers at IEEE INFOCOM '24 and VTC '24; passionate about efficient, reliable networked systems.

Skills

Programming Languages: C++, Python(Pytorch, Numpy, Matplotlib), Bash/Shell, HTML

Technologies & Tools: TCP/IP, NS-3, PyCharm, Docker, CI/CD, Git, SQL, Linux & Windows

Work Experience

Research Assistant - University of Victoria — Jul. 2025 - Current

- Built a deep-learning-based model for satellite signal-strength prediction from geospatial data.

Research Assistant - University of Victoria — Sep. 2021 - Aug. 2024

- Prototyped protocol components for next-generation networks in C++/Python using Network Simulator 3 (ns-3), improving efficiency and enabling new applications.

Teaching Assistant ECE363 - University of Victoria — Jan. 2022 - Apr. 2023

- Designed and guided lab projects on network analysis. Mentored students on network project implementation.

Research Assistant - Institute of Computing Technology, CAS — Aug. 2019 - Dec. 2019

- Optimized server resource scheduling using heuristic algorithms (ant colony, genetic, and simulated annealing) to maximize resource utilization.

Projects

Delay-guaranteed Routing Protocol — Github

Stack : C++, Python, ns-3, Linux, TCP/IP

- Designed and developed DDR, a novel distributed, traffic-aware routing protocol to provide packet-level delay guarantees for time-sensitive applications.
- Automated Linux/ns-3 experiments on ISP-grade graphs; traffic mixes covered mice/elephant/bursty flows; compared against ECMP/LFID/DGRP.
- Impact: reached near-100 % deadline hit rate under modest latency targets; under heavy/bursty load, DDR kept high on-time delivery while baselines degraded; initialization cost stayed lightweight (compute/memory only a small multiple of OSPF).

Multipath Routing Compatible TCP — Github

Stack : C++, Python, ns-3, Linux, TCP/IP

- Implemented an RTT-aware FastRTO to decouple reordering from loss, reducing false fast-retransmits on multipath.
- Built reproducible ns-3/Ubuntu experiments; benchmarked vs NewReno/CUBIC/BBR on OSPF (single-path) & ECMP (multipath) with real ISP topologies.
- Impact: on ECMP multipath, sustained 20–30 Mbps and 3–10× higher throughput than baselines; single-path parity with NewReno/CUBIC/BBR.

Education

University of Victoria — Sep. 2021 – Aug. 2024

Master of Applied Science - Electrical and Computer Engineering
CNLab (Computer Network Lab), Advisor: Lin Cai

Beijing University of Posts and Telecommunications — Sep. 2015 – Jun. 2020

Bachelor of Engineering - Telecommunication Engineering

Publications

Multipath Routing Compatible Congestion Control VTC 2024
Tianfang Chang, Lin Cai
2024 IEEE 100th Vehicular Technology Conference

DDR: A Deadline-Driven Routing Protocol for Delay Guaranteed Service INFOCOM 2024
Pu Yang, **Tianfang Chang**, Lin Cai
2024 IEEE International Conference on Computer Communications

DGR: Delay-Guaranteed Routing Protocol MASS 2023
Pu Yang, Lin Cai, **Tianfang Chang**
2023 IEEE 20th International Conference on Mobile Ad Hoc and Smart Systems

Honors & Awards

Fellowship, University of Victoria Graduate Student 2021-2024
Finalist, IEEE VTS Storytelling Competition 2023
Finalist, IEEE 2023 Workshop on Ubiquitous Poster Competition 2023
Scholarship, University of Victoria Graduate Student 2022

Community Activity

Paper Reviewer 2022-2024
IEEE INFOCOM, Network Magazine, TWC, TMC

Volunteer 2023
Volunteer of IEEE 2023 Workshop on Ubiquitous

Volunteer 2023
Volunteer of Explore UVIC Event